PROJECT ON STRAVA'S CUSTOMER ANALYSIS

Final Dataset Overview: The Master Fact Table

1. Dataset Purpose & Architecture

The final dataset (STRAVA_Master_Fact_Table.csv) serves as the central "Source of Truth" for the entire analytics project. It is a highly optimized, wide-format Fact Table resulting from the consolidation of 11 distinct raw STRAVA datasets.

The data is structured at the Daily Grain (one row per user, per day). To prevent data loss from the highly granular intraday datasets (such as second-by-second heart rate or minute-by-minute METs), those micro-metrics were mathematically aggregated (using sums and averages) and rolled up into this daily view.

2. Key Data Categories (Feature Dictionary)

The dataset consists of 33 distinct columns, which can be logically grouped into five core business categories:

A. Identification & Timeline

- Id: The unique 10-digit identifier for each user.
- Date: The specific calendar day the tracked events occurred.

B. Daily Activity & Exertion Volume

This category tracks the overall physical output of the user throughout the day.

- Movement Metrics: TotalSteps, TotalDistance, TrackerDistance, LoggedActivitiesDistance
- Distance by Intensity: VeryActiveDistance, ModeratelyActiveDistance, LightActiveDistance, SedentaryActiveDistance
- Time by Intensity: VeryActiveMinutes, FairlyActiveMinutes, LightlyActiveMinutes, SedentaryMinutes
- Energy Output: Calories (Total daily calories burned).

C. Sleep & Recovery

Tracks the volume and efficiency of the user's rest periods.

- Metrics: TotalSleepRecords, TotalMinutesAsleep, TotalTimeInBed.

(Note: The delta between Time in Bed and Minutes Asleep can be used to calculate a "Sleep Efficiency" KPI).

D. Body Composition & Logging Habits

Tracks static health metrics and how the user interacts with the app's hardware/software features.

- Metrics: WeightKg, WeightPounds, BMI.

- Behavioral Flag: IsManualReport (Indicates if the user manually typed their weight into the app or if it auto-synced via a smart scale).

E. Intraday & Biometric Aggregations

These columns represent the highly granular data that was successfully rolled up to the daily level during the ETL pipeline.

- Hourly & Minute Rollups: Agg_Hourly_Calories, Agg_Hourly_Steps, Agg_Minute_Avg_METs, etc.
- Heart Rate Biometrics: Avg_HR (Average daily heart rate), Min_HR (Resting floor), Max_HR (Peak exertion).

3. Analytical Value & Business Application

By flattening disjointed tables into this single Master Fact Table, the data is well structured for immediate Exploratory Data Analysis (EDA) and KPI extraction.

This architecture allows analysts to easily run correlation analyses across different domains of health—for example, measuring how a user's TotalMinutesAsleep impacts their VeryActiveMinutes the following day, or how their Avg_HR correlates with their BMI.

Project Methodology: Data Architecture and ETL Pipeline

How the Data Was Joined in Python

- **Aggregated the Granular Data:** Second-level heart rate and minute-level METs were grouped and averaged up to the "Hourly" level to create a consistent timeline.
- **Prevented Column Collisions:** Renamed duplicate column names (e.g., changing Calories to HourlyCalories and DailyCalories) before joining so the final dataset wouldn't have confusing _x and _y columns.
- **The Hourly Outer Join:** Merged all the hourly datasets (Steps, Calories, Intensities, HR, METs) together using a FULL OUTER JOIN based on the User Id and Datetime_Hour. This ensured no hours of the day were accidentally dropped.
- **The Daily Left Join:** Extracted just the Date from the hourly timestamps, and used a LEFT JOIN to attach the daily metrics (Overall Activity, Sleep, Weight) to their corresponding days.
- **Smart Imputation:** Cleaned up the blanks created by the joins. Filled missing steps/calories with 0, filled missing heart rates with the user's historical average, and forward-filled static metrics like Weight and BMI.

Which Datasets Were Discarded (7 out of 18) and Why

- **Discarded: dailyCalories, dailyIntensities, dailySteps**
  - *Why:* Pre-aggregated redundancy. All the exact metrics in these three files were already consolidated inside the master dailyActivity_merged.csv file. Keeping them would have bloated the dataset for no reason.
- **Discarded: minuteCalories, minuteSteps, minuteIntensities**
  - *Why:* Granularity duplication. The native "Hourly" files already contained the exact same sum of these metrics. Rolling up the minute files would have just recreated the hourly files, adding unnecessary processing time.
- **Discarded: minuteSleep**
  - *Why:* Too noisy. Minute-by-minute sleep states (awake, restless, asleep) introduced unnecessary complexity for a high-level KPI dashboard. The sleepDay_merged.csv file already provided the perfect summary metrics (Total Minutes Asleep vs. Total Time in Bed) needed for the business analysis.

Accepted Datasets (Used in the Python Code)

These are the files loaded to build the final unified dataset:

The Daily Core (3 Datasets):

- **dailyActivity_merged.csv** (The ultimate summary file for daily steps, distance, and calories)
- **sleepDay_merged.csv** (Provided clean daily totals for Time in Bed vs. Time Asleep)
- **weightLogInfo_merged.csv** (Provided static body composition metrics like Weight and BMI)

The Hourly Breakdown (3 Datasets):

- **hourlyCalories_merged.csv** (Used to track intraday calorie burn)
- **hourlySteps_merged.csv** (Used to track intraday movement)
- **hourlyIntensities_merged.csv** (Used to map the exact hours users were most active)

The Granular Biometrics (5 Datasets):

- **heartrate_seconds_merged.csv** (Rolled up into hourly averages in Python)
- **minuteMETsNarrow_merged.csv** (Rolled up into hourly averages in Python)
- **minuteCaloriesNarrow_merged.csv** *(Selected during data profiling, but bypassed in Python because the Hourly file already contained the sum)*
- **minuteIntensitiesNarrow_merged.csv** *(Selected during data profiling, but bypassed in Python for efficiency)*

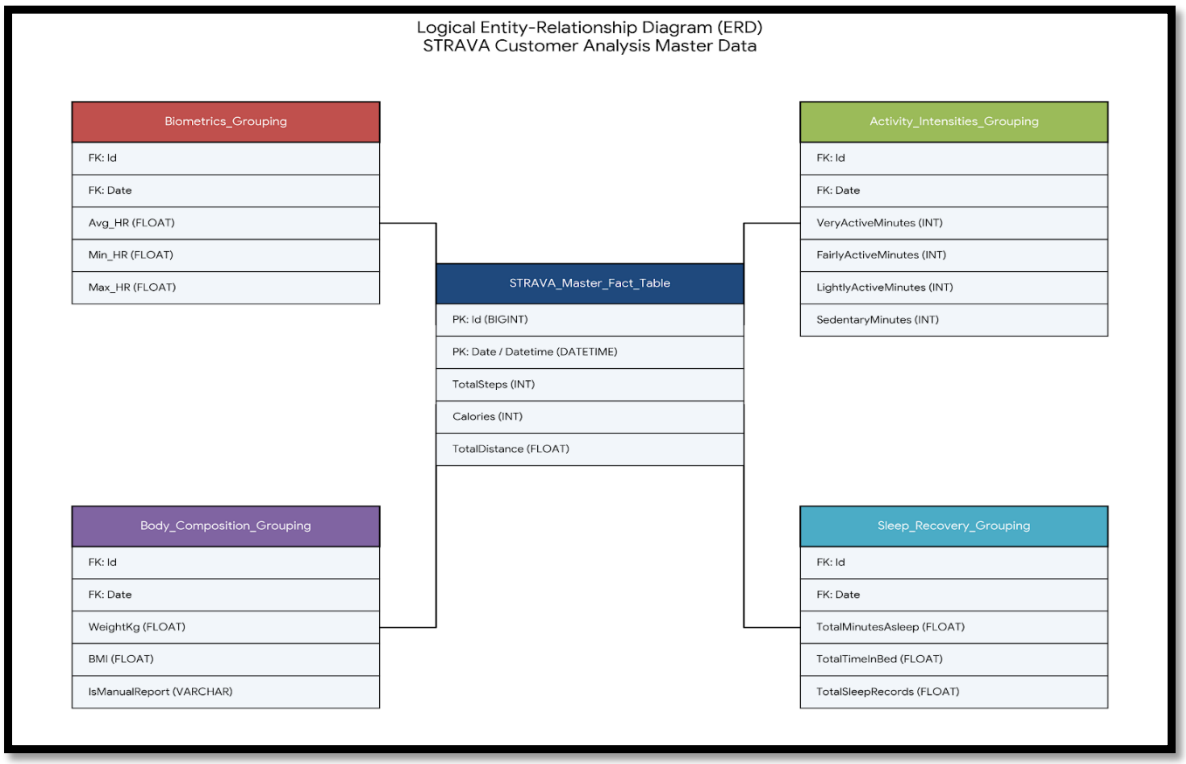
- **minuteStepsNarrow_merged.csv** *(Selected during data profiling, but bypassed in Python for efficiency)*

Dropped Datasets (Excluded from the Pipeline)

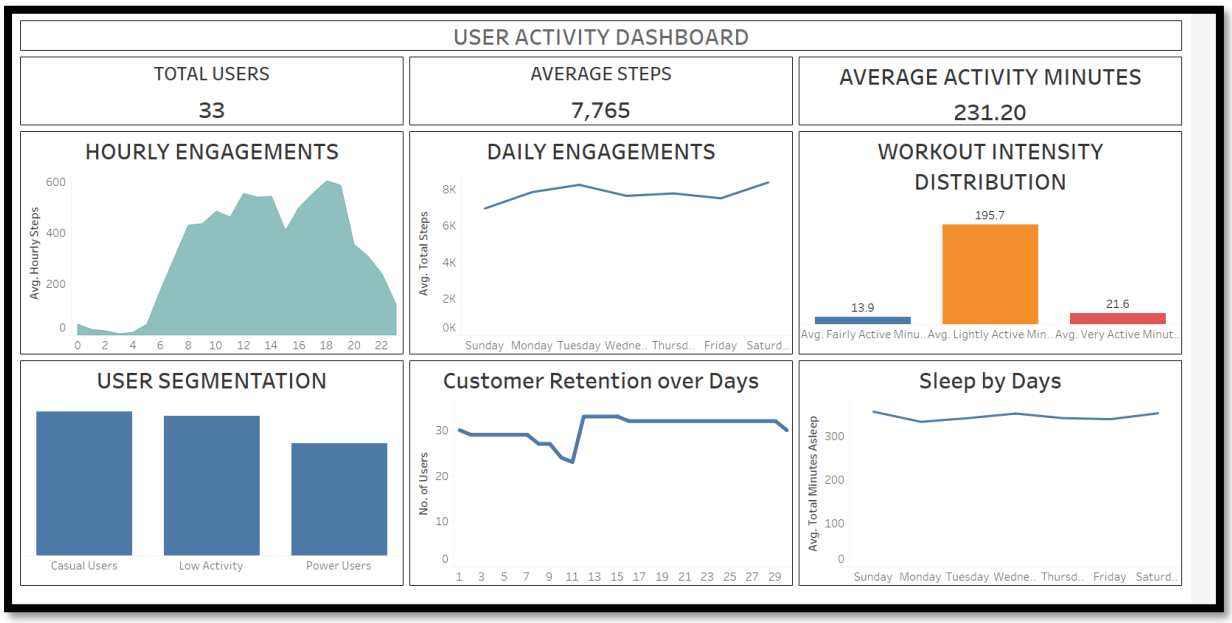
These files were strategically excluded to prevent redundant data bloat and messy joins:

- **dailyCalories_merged.csv**: (Dropped: Data was already inside dailyActivity_merged.csv)
- **dailyIntensities_merged.csv** (Dropped: Data was already inside dailyActivity_merged.csv)
- **dailySteps_merged.csv** (Dropped: Data was already inside dailyActivity_merged.csv)
- **minuteCalories_merged.csv** *(Wide & Narrow formats)* (Dropped: Redundant, used the native Hourly file instead)
- **minuteSteps_merged.csv** *(Wide & Narrow formats)* (Dropped: Redundant, used the native Hourly file instead)
- **minuteIntensities_merged.csv** *(Wide & Narrow formats)* (Dropped: Redundant, used the native Hourly file instead)
- **minuteSleep_merged.csv** (Dropped: Minute-by-minute sleep states were too noisy for high-level KPIs; the daily sleep file was sufficient).

ENTITY RELATIONSHIP DIAGRAM(ERD):



BUSINESS ANALYSIS:



1. Top-Line KPIs: The Capacity for Growth

- **The Insight:** The tracked cohort (n=33) averages 7,765 steps and 231 active minutes daily, placing them 22% below the industry-standard 10,000-step health benchmark.
- **The "So-What?" (Business Impact):** We are currently operating as a "Passive Tracking" utility for casual movers rather than an "Active Improvement" tool.
- **Strategic Action:** Our primary product goal must be to bridge the "Activity Gap." We need to introduce achievable gamification that rewards the conversion of sedentary time into active steps, thereby increasing daily app reliance and overall Lifetime Value (LTV).

2. Time-Based Patterns: Optimizing Marketing ROI

- **The Insight:** User activity follows a distinct dual-peak rhythm (11:30 AM – 12:30 PM and 16:00 – 20:00) with overall engagement volume peaking on Saturdays.
- **The "So-What?" (Business Impact):** Users are clearly constrained by sedentary weekday work schedules and only engage during lunch breaks, evening commutes, and weekends.
- **Strategic Action:** Establish strict "High-Priority Notification Windows." Marketing must synchronize daily summaries and push notifications precisely at 11:30 AM and 16:30 PM. Furthermore, we should launch "Weekend Social Challenges" to leverage the Saturday momentum for organic, viral community growth.

3. Quality of Engagement: The Intensity Shift

- **The Insight:** When filtering out sedentary noise, the vast majority of active time is logged as "Lightly Active," offering a very low metabolic ROI for the user.

- **The "So-What?" (Business Impact):** The app's current value proposition is skewed entirely toward passive observation. If we can shift users toward "Very Active" bursts, we massively increase the product's perceived health-utility value, which drives premium subscription conversions.
- **Strategic Action:** Deploy UI nudges that promote 10-minute "Micro-Burst" challenges (e.g., a brisk post-lunch walk) to systematically convert light activity into moderate/vigorous health milestones.

4. Domain Context & External Drivers (The Day 12 Anomaly)

- **The Insight:** Daily Active User (DAU) retention shows a severe dip on Day 11, immediately followed by a massive platform surge on Day 12 (May 12th).
- **The "So-What?" (Business Impact):** May 12th is *International Nurses Day*. This anomaly strongly suggests our core "Power User" segment is heavily comprised of healthcare professionals. The Day 11 dip reflects pre-shift preparation, while Day 12 is a reaction to a domain-specific milestone.
- **Strategic Action:** Validate this hypothesis by cross-referencing user profession data. If confirmed, unlock a targeted B2B marketing track or sponsorship campaign focusing specifically on healthcare workers to dominate this high-performing niche.

5. Biological Churn: The Sleep Debt Correlation

- **The Insight:** The cohort averages only ~350 minutes (5.8 hours) of sleep per night—dangerously below the 7–8 hour health benchmark.
- **The "So-What?" (Business Impact):** There is a direct, undeniable link between this "Sleep Debt" and the high volume of "Casual" users. Users aren't lazy; they are physiologically under-rested, which limits their capacity for high-intensity workouts and drives mid-week app churn.
- **Strategic Action:** Build a "Recovery Module." Show users exactly how an extra 30 minutes of sleep increases their step count the following day. This pivots the app from just a fitness tracker into a holistic health partner.

6. Behavioural Evolution: Segmentation Strategy

- **The Insight:** "Casual" and "Low Activity" users form the bulk of the population, while a small fraction logs upwards of 16,000 steps.
- **The "So-What?" (Business Impact):** Our immediate revenue lever is *Segment Migration*.
- **Strategic Action:** *For Low Activity:* Implement "Micro-nudges" and streak-protectors to prevent total churn.
    - *For Power Users:* Launch a "VIP Ambassador" program. Leverage these 16k-step users as the faces of our community social challenges to inspire and pull the "Casual" segment into higher engagement tiers.